Whole Genome CRISPR-cas9 Screening on Isogenic Cell Line Model Systems to Identify Synthetic Lethal Partners of the Glioma Associated CIC Transcriptional Repressor

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Introduction

The capicua gene (*CIC*) encodes a transcriptional repressor that is involved in the development of the central nervous system and is observed to harbour mutations in 70% of patients with Type 1 Low Grade Glioma (LGG). These somatic mutations are hypothesized to inactivate the CIC protein, implicating a tumour suppressor role, though CIC's involvement in tumour progression is not well understood.

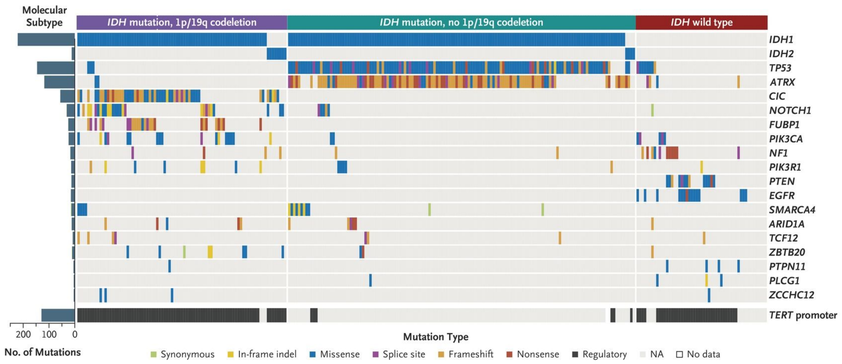


Figure 1. Mutational landscape of somatic mutations in lower-grade gliomas (Cancer Genome Atlas Network, 2015). Mutations in *CIC* appear to co-occur with *IDH1/2* mutations and 1p/19q co-deletions and appear to be mutually exclusive with mutations in *ATRX* and *TP53*.

If inactivating two genes simultaneously results in death while inactivating one alone does not, the gene pair are said to be **synthetic lethal partners**. Genes that are synthetic lethal partners are functionally redundant such that one is able to compensate for the loss of the other.

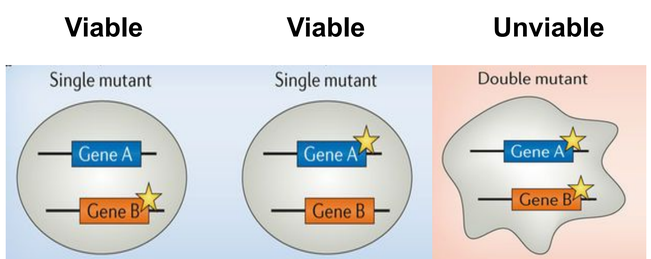


Figure 2. The concept of synthetic lethality. Since the functions of genes A and B are collectively required for viability, one may infer that they participate in parallel pathways to keep the cell alive. (Adapted from O'Neil et al, 2017)

Objective

We will perform a whole genome CRISPR-cas9 screen to identify *CIC* synthetic lethal partners to better understand what cellular pathways CIC may be involved in.

Methods

Isogenic *CIC*-wildtype (WT) and *CIC*-knockout (KO) HEK293 and glioma derived HOG and BT054 cell lines will be generated using CRISPR-cas9. To conduct the whole genome CRISPR-Cas9 screen, we will use a single-guide RNA (sgRNA) library that will contain 5 or more guides per gene in the genome. Given the definition of synthetic lethality, we expect that guides targeting *CIC* synthetic lethal partners will be depleted in *CIC*-KO cells but not in *CIC*-WT cells.

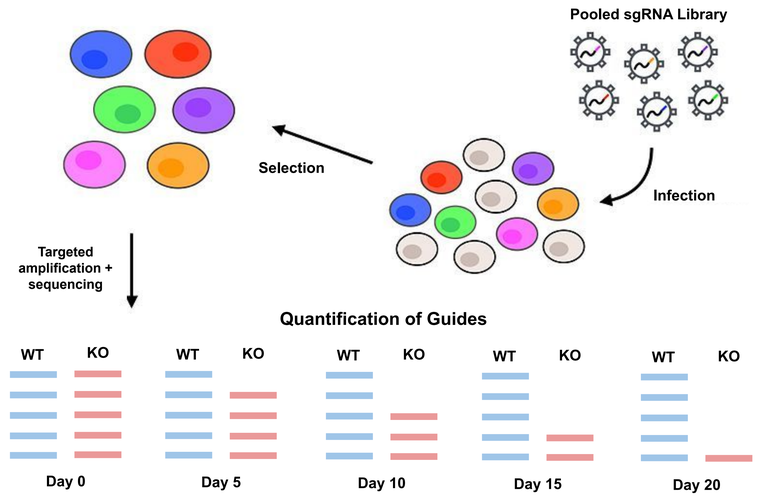


Figure 3. Overview of the whole-genome CRISPR-Cas9 synthetic lethal screen (adapted from https://en.wikipedia.org/wiki/Perturb-seq). The sgRNA library is delivered via lentiviral particles at low concentration to ensure that cells will not receive more than a single guide. Selection using an antibiotic is performed to enrich for successfully infected cells. Genomic DNA is isolated every 5 days post infection and PCR amplified at the region containing a molecular barcode that serves to uniquely identify each guide. Sequencing and analysis to quantify the abundance of guides then takes place.

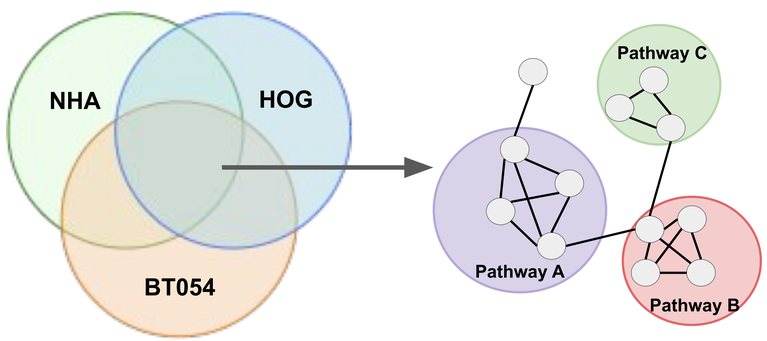


Figure 4. Selection of top synthetic lethal candidates and downstream analysis. *CIC* synthetic lethal partners that are identified in all three isogenic cell lines will constitute our top candidates. These genes will undergo pathway enrichment analysis to identify what biological processes are represented.

Expected Results & Discussion

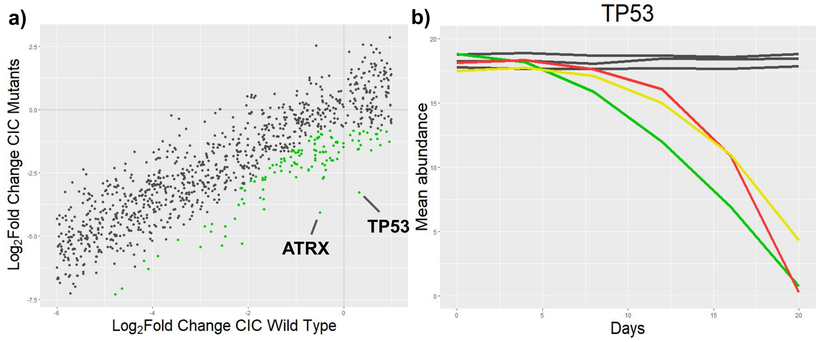


Figure 4. a) Fold changes of guides averaged for each gene between day 0 & day 20. Green dots indicate *CIC* synthetic lethal partner candidates. **b)** Mean abundance of guides targeting TP53 over 20 days. Coloured lines represent NHA, HOG and BT054 *CIC*-KO cell lines and black lines represent all the *CIC*-WT cell lines.

We expect that guides targeting *TP53* and *ATRX* will exhibit a dropout in *CIC*-KO cells as displayed above. *TP53* and *ATRX* are suspected *CIC* synthetic lethal partners due to the mutual exclusivity of mutations between *CIC* and these genes (figure 1). TP53 is a critical regulator of the cell cycle that will initiate pro-apoptotic signals upon detecting genomic instability (Aubrey et al, 2016) while ATRX functions as an epigenetic regulator of chromatin structure and its loss has been associated with chromosome instability (Ritchie et al, 2008). Therefore, synthetic lethal interactions between *CIC* and *ATRX* or *TP53* would potentially highlight their functional convergence on maintaining genomic integrity.

Conclusions & Future Directions

- TP53 and ATRX are among the expected top *CIC* synthetic lethal partner candi\dates
- Genes related to *TP53* and *ATRX* are also expected to exhibit synthetic lethality with *CIC*
- To validate *CIC* synthetic lethal interactions, we will assess for cellular viability after treatment with small interfering RNAs targeting our top candidates
- It is expected that these findings will contribute to our understanding regarding CIC's role in tumorigenesis and, therefore, potentially unveil novel opportunities for therapy specific for *CIC*-mutant tumours

References

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